DTW + 1-NN — Non-Parametric Reference

Per-model deep dive · the zero-parameter memoriser that beats every trained single model

Result: 0.567 accuracy · 0.572 weighted F1 · WER 0.433 · fold SD ±0.072

Method: Dynamic Time Warping distance + 1-nearest-neighbour (added as the honest baseline)

1 Theory

Dynamic Time Warping measures how similar two sequences are while allowing them to be stretched or compressed in time — so the same sign performed slowly or quickly still matches. Paired with **1-nearest-neighbour**, classification is simply: find the most similar training video and copy its label. It fits **zero parameters** — it stores the training data and compares. It makes essentially one assumption (two performances of the same sign can be aligned in time), so there is almost nothing for it to get wrong.

There is a neat connection to the HMM: DTW is roughly Viterbi decoding on a degenerate HMM where every frame of a stored template is its own state. Where the HMM compresses 35 frames into 3 states, DTW keeps all 35 — maximum states, zero estimation.

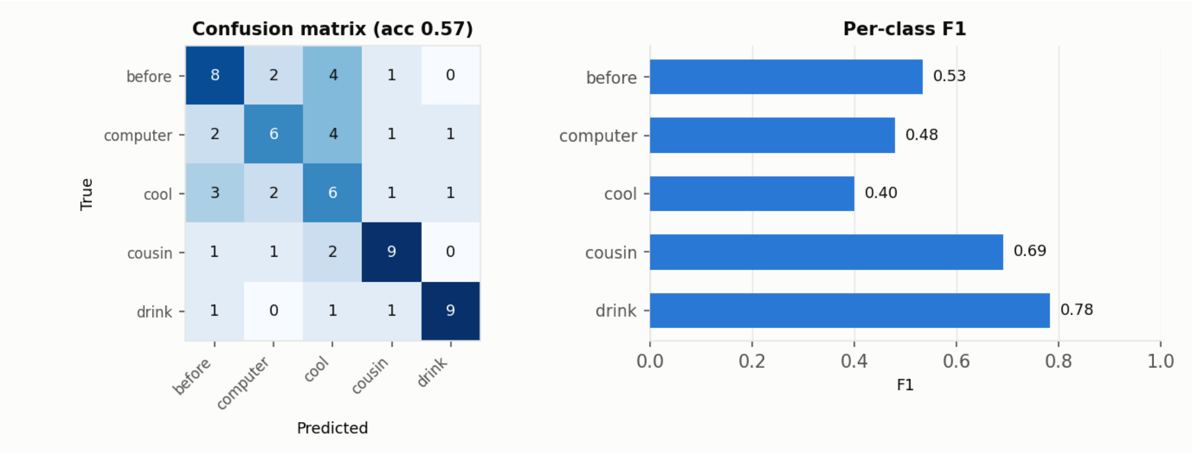
2 Method of execution

Each video is reduced to the same 16-D PCA landmark features used by the corrected HMM. A test video's DTW distance to every training video is computed via the standard dynamic-programming recurrence, and the nearest training video's label is assigned. It is evaluated on the identical folds as every other model, so the comparison is exact.

```
D[i,j] = dist(a_i, b_j) + min(D[i-1,j], D[i,j-1], D[i-1,j-1])   # DTW recurrence
predict(test) = label of argmin over training videos of DTW(test, train)
```

3 Results

Metric	Value (5-fold CV)
Accuracy	0.567
Weighted precision	0.583
Weighted recall	0.567
Weighted F1	0.572
Word Error Rate (WER)	0.433
Fold accuracies	0.57, 0.43, 0.62, 0.62, 0.62
Cross-val SD	±0.072
Distinct classes predicted	5 of 5
Majority baseline	0.224



DTW + 1-NN confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

DTW is the **strongest single method** at 0.567, beating every trained architecture. This is the textbook bias–variance trade-off: with ~13 examples per class, every parameter a model estimates carries error, and those errors compound. DTW estimates *nothing*, so it has no estimation error to accumulate — the rigid method wins at small sample size. That two structurally unrelated methods (DTW and the HMM) agree closely on which classes are hard is itself evidence that the difficulty lives in the data, not the model.

What makes it stronger / weaker

Stronger: unbeatable at tiny scale, no assumptions to violate, trivial to extend with new classes. **Weaker:** it never generalizes or compresses — prediction cost grows with the dataset (every test video compared to all training videos) and it gives distances, not calibrated probabilities. Unusable in real time at 2,000-gloss scale.

5 Role in the hybrid

DTW is the **honest baseline and low-data fallback**, not a component of the trained hybrid. Its importance is as the bar to beat: any trained model — including the hybrid — must exceed DTW to justify its parameters. The two-stream hybrid is the **only** model that clears it, which is what makes the hybrid's result meaningful. Keep DTW in the study as the reference against which real progress is measured, and expect trained models to overtake it decisively only once the dataset reaches ~30–50 videos per class.